

IG BLOCK and CAPL



AGENDA

- Introduction
- IG Block { What are they?
Why are they useful?
- CAPL { HW & CANalyzer's configuration
- Examples
- Q&A
- Exercises



INTRODUCTION

During V&V activities on both vehicle and HIL, sometimes it is necessary to change signals that ECUs send or receive on CAN bus. In some cases, the maneuvers to vary these signals aren't suitable to be replicated physically due to:

- Dangerous conditions for the driver
- Damages to vehicle components (sensors, actuators, wirings, ECUs)
- Difficult vehicular conditions
- Long-time or particular testing conditions

Others application cases could be:

- Loss of Communication (LoC) between real ECUs
- MC (Message Counter) and CRC (Cyclic Redundancy Check) errors (Implausibility Faults)

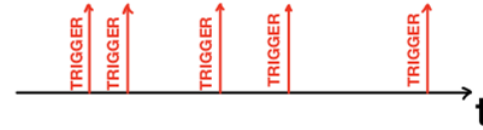
IG BLOCK and CAPL can be used for all those purposes



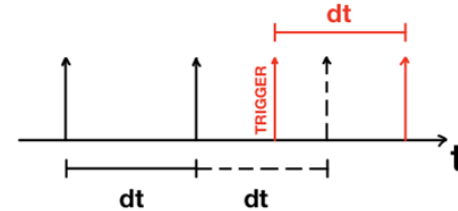
INTRODUCTION

On CAN Bus, different types of signals can be sent and received from ECUs:

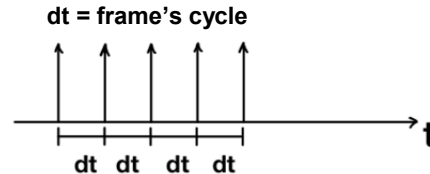
- On Event (e.g. Diagnostic Frames)



- Cyclic On Event (e.g. Impact Frames)



- Cyclic

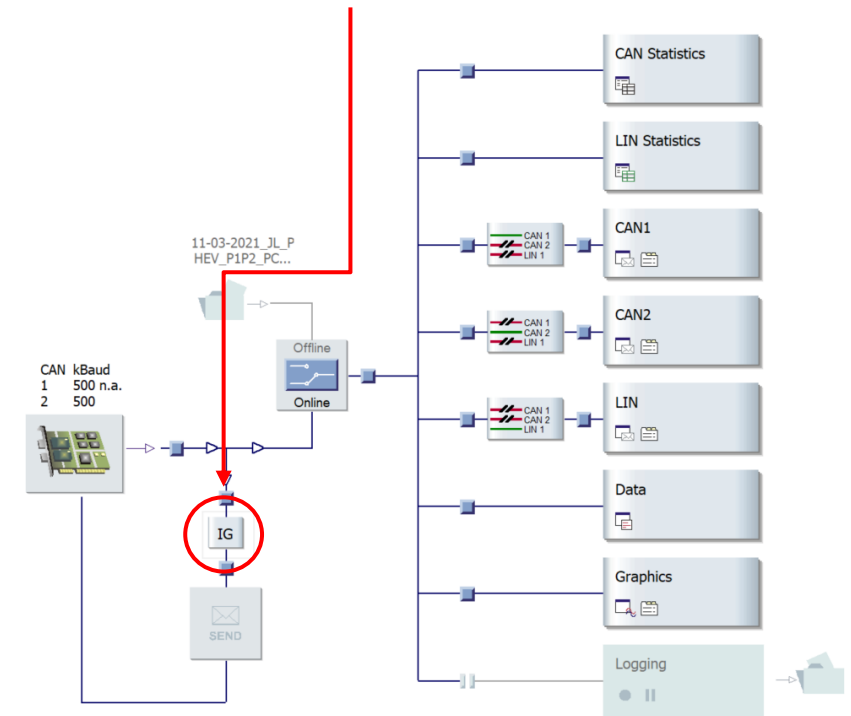


According to signals' type, we can choose which type of gateway to be used

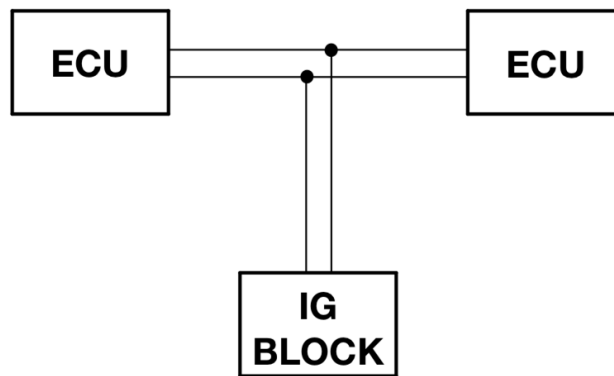


IG BLOCK

The Interactive Generator Block (IG) is used to create and send **On Event** messages ONLY.



WIRINGS' SCHEME: T-CONNECTION
The IG node is in parallel with the CAN bus





IG BLOCK

For this purpose, both a list of messages and trigger conditions are the input in a configuration dialog. All the signals included in the selected messages are listed too and their values can be freely changed, according to the associated CAN database (DBC of the architecture's project under testing). The main characteristic of the Interactive Generator Block is the possibility to configure and interactively send messages during a running measurement (online).

Signals of the selected message are shown in the signal list

Transmitted frame list

Message Name	Channel	DLC	Send	Triggering	Data Field
				Cycle Time [ms]	0 1 2 3 4 5 6 7
TrainingNetwork_fmSensorResult	CAN 1	3	now	1	0 0 0
TrainingNetwork_fmKey8	CAN 1	4	now	10	0 0 0 0
TrainingNetwork_fmKey12	CAN 1	1	now	10	0
TrainingNetwork_fmLamps	CAN 1	1	now	123	0
TrainingNetwork_fmSensorA	CAN 1	8	now	250	0 0 0 0 0 0 0 0
TrainingNetwork_fmSensorB	CAN 1	8	now	250	0 0 0 0 0 0 0 0
TrainingNetwork_fmStatus	CAN 1	4	now	200	0 0 0 0 0 0

Signal list

SB	Signal Name	Raw Value	Phys Value	Unit	Dec	Phys Step	Inc	Wave form gen
8	sigRotKnob	0	0	%	-	4.985372434	+	None
7	sigKey8	☒	Key pressed		-	1	+	None
6	sigKey7	☒	Key pressed		-	1	+	None
5	sigKey6	☐	Key not pressed		-	1	+	None
4	sigKey5	☒	Key pressed		-	1	+	None
3	sigKey4	☐	Key not pressed		-	1	+	None
2	sigKey3	☐	Key not pressed		-	1	+	None
1	sigKey2	☐	Key not pressed		-	1	+	None
0	sigKey1	☐	Key not pressed		-	1	+	None



IG BLOCK

- For each message, its parameters such as ID and DLC can be seen in the configuration window
- The trigger conditions (Triggering) specify the instant of time the message will be sent
- The content of the message (i.e. signals) can be changed in the Data Field (hex values) or in the Signal list related to the frame (hex values or physical values)

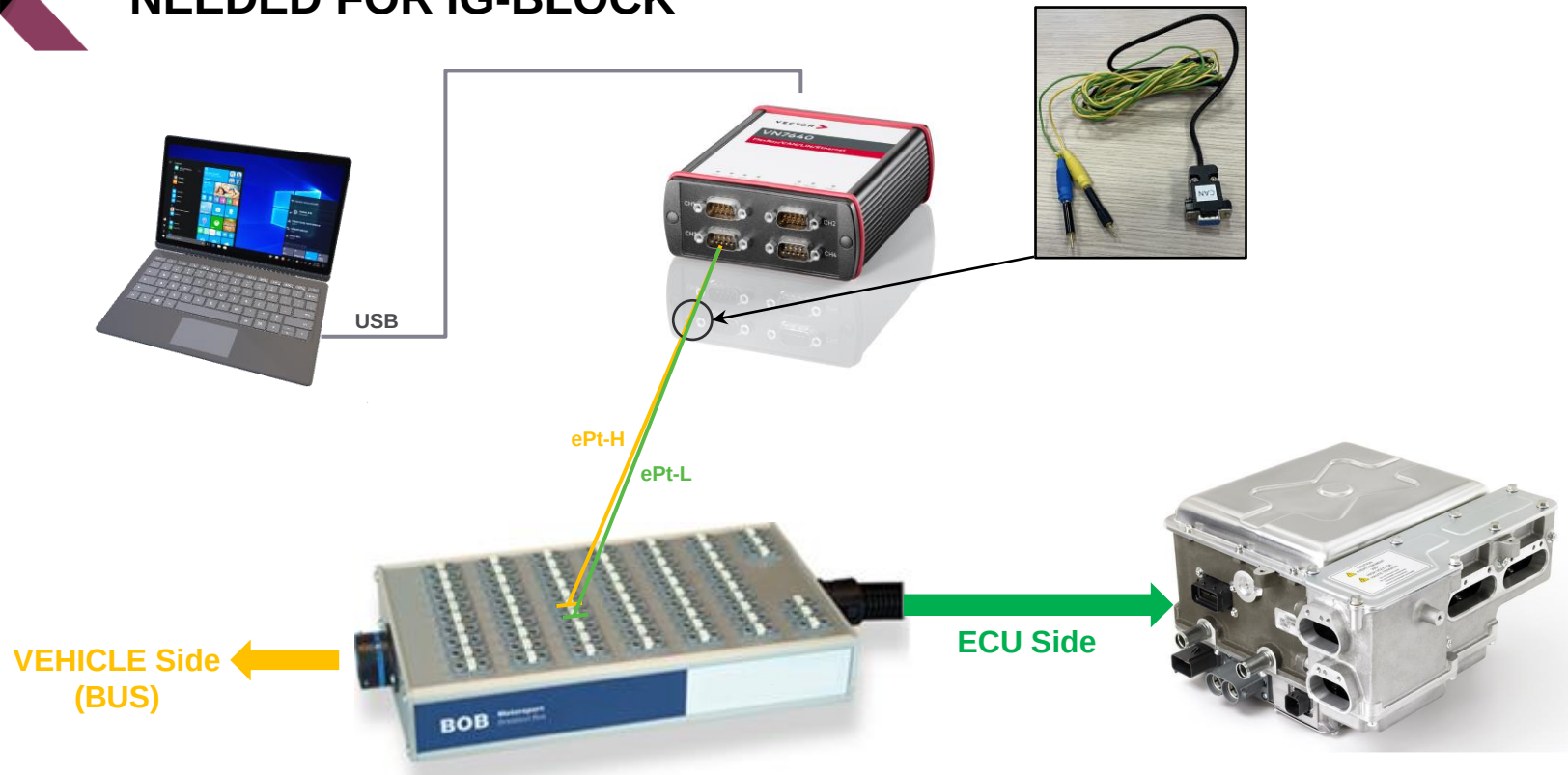
SB	Signal Name	Raw Value	Phys Value	Unit	Dec	Phys Step	Inc	Wave form generation		Modification
168	ThrottlePosition	0	0	%	-	5.096	+	None	Define...	☐ enabled
80	RawPVS2	0	0	V	-	0.25490195	+	None	Define...	☐ enabled
88	RawPVS1	0	0	V	-	0.25490195	+	None	Define...	☐ enabled
152	ACPressure	1FF	SNA	bar	-	2	+	None	Define...	☐ enabled
104	VehicleSetSpeed	0	0	km/h(mph)	-	13	+	None	Define...	☐ enabled
251	TransWarmUpReq	☐ 0	WarmUp_Not_E	-	-	1	+	None	Define...	☐ enabled
96	TransWarmUpMin	0	0	rpm	-	325	+	None	Define...	☐ enabled
249	ThermalManagem	☐ 0	Not_active	-	-	1	+	None	Define...	☐ enabled
200	RawTPS2	FF	4.999999965	V	-	0.25490195	+	None	Define...	☐ enabled
192	RawTPS1	FF	4.999999965	V	-	0.25490195	+	None	Define...	☐ enabled
72	MessageCounter_E	0	0	-	-	1	+	None	Define...	☐ enabled
255	Lv1CrsCtrlOff	☐ 0	0	-	-	1	+	None	Define...	☐ enabled
254	ESP_TrqRqEnabl	☐ 0	0	-	-	1	+	None	Define...	☐ enabled
128	EngMaxRPM_W	0	0	-	-	500	+	None	Define...	☐ enabled
64	CRC_ENGINE1	0	0	-	-	13	+	None	Define...	☐ enabled

Default input mode: CAN

Close Help



HW CONFIGURATION NEEDED FOR IG-BLOCK



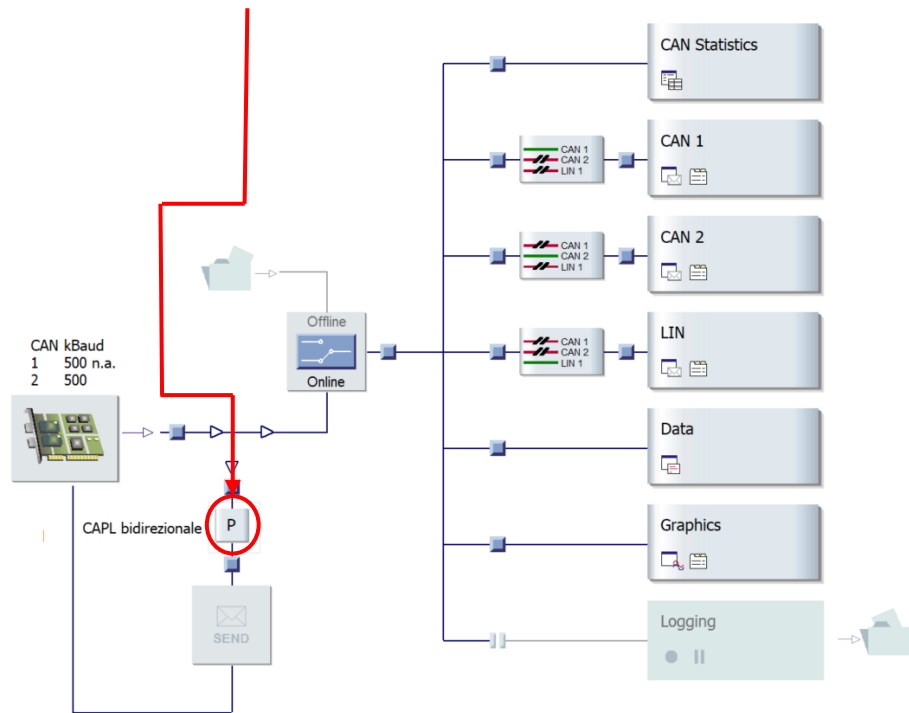
VEHICLE Side
(BUS)

ECU Side

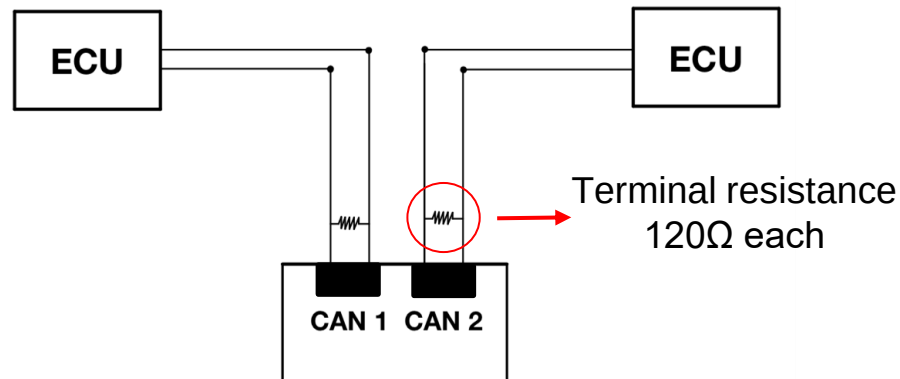


CAPL

The CAPL is used to generate and send Cyclic and Cyclic On Event messages.



WIRINGS' SCHEME: S-CONNECTION
We need to physically interrupt the CAN bus





1. Basic CAPL code : used to overturn messages from a port to the other (no frame modification)

↓ ↑

```
on message CAN2.*           //Focus on CAN2 port on CANcase
{
    message CAN1.* msg;      //Take the message on CAN1 port which is passing through the channel

    if(this.dir != rx) return; //If this message is transmitted by CAN1 port out from CANcase, exit from the function
    output(msg);              //If the message is received by CAN1 port, send it out from CANcase using CAN2 port

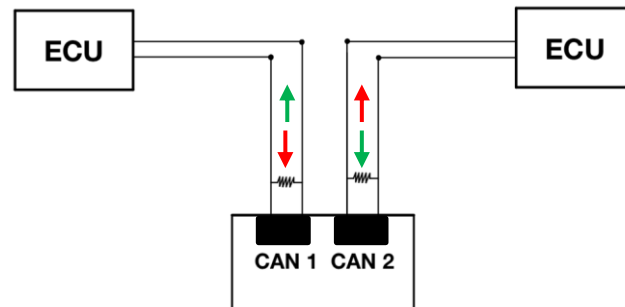
    return;
}
```

↓ ↑

```
on message CAN1.*           //Focus on CAN1 port on CANcase
{
    message CAN2.* msg;      //Take the message on CAN2 port which is passing through the channel

    if(this.dir != rx) return; //If this message is transmitted by CAN2 port out from CANcase, exit from the function
    output(msg);              //If the message is received by CAN2 port, send it out from CANcase using CAN1 port

    return;
}
```





CAPL

2. Advanced CAPL code : used to change the signal's value of a specific message

```
2 variables
3 {
4   int i, i_byte, i_bit, crc_reg, X, sign=0;
5   dword crc_resto, poly;
6   int array_bytes[8];
7 }
8
9
10 on key '1' //Pressing button "1" on keyboard we set variable X equal to 1
11 {
12   X = 1;
13 }
14 on key '0' //Pressing button "0" on keyboard we set variable X equal to 0 (used for reset)
15 {
16   X = 0;
17 }
18
19 on message CAN2.* //Focus on CAN2 port on CANcase
20 {
21
22   message CAN1.* msg; //Take the message on CAN1 port which is passing through the channel
23
24   if(this.dir != rx) return; //If this message is transmitted by CAN1 port out from CANcase, exit from the function
25
26   if(this.id==0x150 & X ==1) //Check if the message ID is equal to 150 hex and X=1
27   {
28     message CAN1.BPCM_MSG_01 msg; //Assign to variable "msg" the message we want to modify
29     msg.HVBatCntctrOpn=1; //Choose signal to be modified inside the message
30     msg.CRC_150h=CRCCalc(msg); //Call the function to calculate the message's CRC
31     output(msg); //Send it in output with changed value
32     msg=this;
33     return;
34   }
35
36   else //If ID is not equal to 150 hex or X!=1 overturn the message only
37   {
38     message CAN1.* msg;
39     msg = this;
40     output(msg);
41     return;
42   }
43 }
```



CAPL

```

2 | {
3 |   int i, i_byte, i_bit, crc_reg, X;
4 |   dword crc_resto, poly;
5 |   int array_bytes[8];
6 | }
7 |
8 | on key '1'
9 | {
10 |   X = 1;
11 | }
12 |
13 | on key '0'
14 | {
15 |   X = 0;
16 | }
17 |
18 | on message CAN2.*
19 | {
20 |
21 |   message CAN1.* msg;
22 |
23 |   if(this.dir != rx) return;
24 |
25 |   if(this.id==0x150 & X ==1)
26 |   {
27 |     message CAN1.BPCM_MSG_01 msg;
28 |     msg.HVBatCntctrOpn=1;
29 |     msg.CRC_150h=CRCCalc(msg);
30 |     output(msg);
31 |     msg = this;
32 |     return;
33 |   }
34 |
35 |   else
36 |   {
37 |     message CAN1.* msg;
38 |     msg = this;
39 |     output(msg);
40 |     return;
41 |   }
42 | }

```

```

45 | on message CAN1.*
46 | {
47 |   if(this.dir != rx) return;
48 |
49 |   if(this.id==0x150 & X ==1)
50 |   {
51 |     message CAN2.BPCM_MSG_01 msg;
52 |     msg.HVBatCntctrOpn=1;
53 |     msg.CRC_150h=CRCCalc(msg);
54 |     output(msg);
55 |     msg = this;
56 |     return;
57 |   }
58 |
59 |   else
60 |   {
61 |     message CAN2.* msg;
62 |     msg = this;
63 |     output(msg);
64 |     return;
65 |   }
66 | }
67 |

```

```

68 | int CRCCalc (message *pippo)
69 | {
70 |   crc_reg=0xFF;
71 |   poly=285;
72 |
73 |   array_bytes[pippo.DLC-1]=(pippo.byte(0) ^ crc_reg);
74 |
75 |   for (i=1;i<pippo.DLC-1;i++)
76 |     array_bytes[pippo.DLC-1-i]=pippo.byte(i);
77 |
78 |   array_bytes[0]=0;
79 |
80 |   crc_resto = (array_bytes[pippo.DLC-1]<<1) + ((array_bytes[pippo.DLC-2] & 0x80) >>7);
81 |
82 |   for (i_byte=pippo.DLC-2;i_byte>=0;i_byte--)
83 |   {
84 |     for (i_bit=7;i_bit>=1;i_bit--)
85 |     {
86 |       if ((crc_resto>>8) & 0x1)
87 |         crc_resto = (crc_resto ^ poly);
88 |
89 |       crc_resto = (crc_resto<<1) + ((array_bytes[i_byte]>>(i_bit-1)) & 0x1);
90 |       if (i_bit==1 && i_byte>0)
91 |       {
92 |         if ((crc_resto>>8) & 0x1)
93 |           crc_resto = (crc_resto ^ poly);
94 |
95 |         crc_resto = (crc_resto<<1) + ((array_bytes[i_byte-1] & 0x80)>>7);
96 |       }
97 |     }
98 |   }
99 |   if ((crc_resto>>8) & 0x1)
100 |     crc_resto = (crc_resto ^ poly);
101 |   crc_resto=(crc_resto) ^ 0xFF;
102 |
103 |   return crc_resto;
104 | }

```

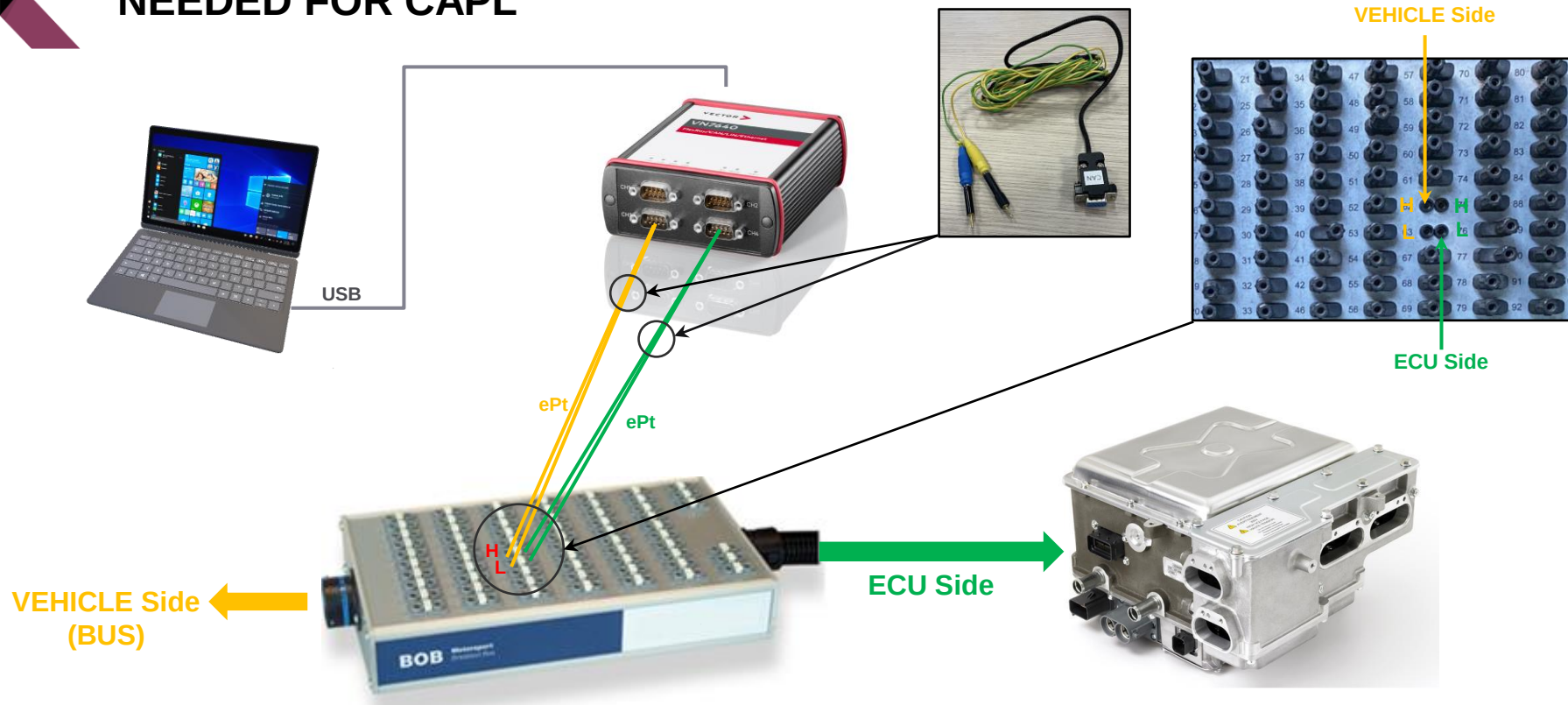


COMMON USES OF CAPL

- To change the signal's value of a specific cyclic and cyclic on event message
- To set a Lost of Communication (LoC) between different ECUs
- To set implausible data transmission/reception (MC, CRC, SNA or validity bit)
- To change more signals of the same CAN bus at the same instant
- To change signals belonging to different CAN buses (e.g. C1, ePT) → Double Gateway

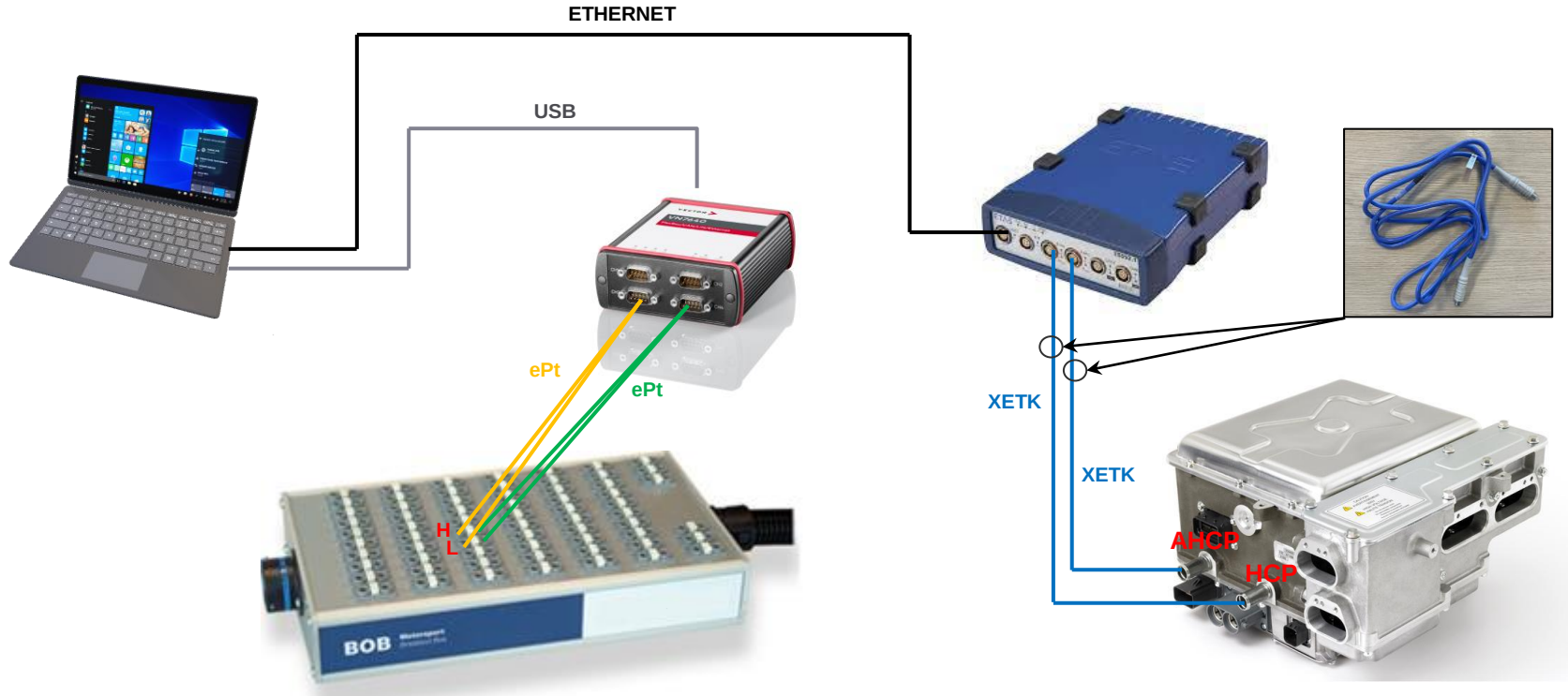


HW CONFIGURATION NEEDED FOR CAPL





SETUP FOR TESTING ACTIVITIES IN THE VEHICLE





CAPL vs IG BLOCK

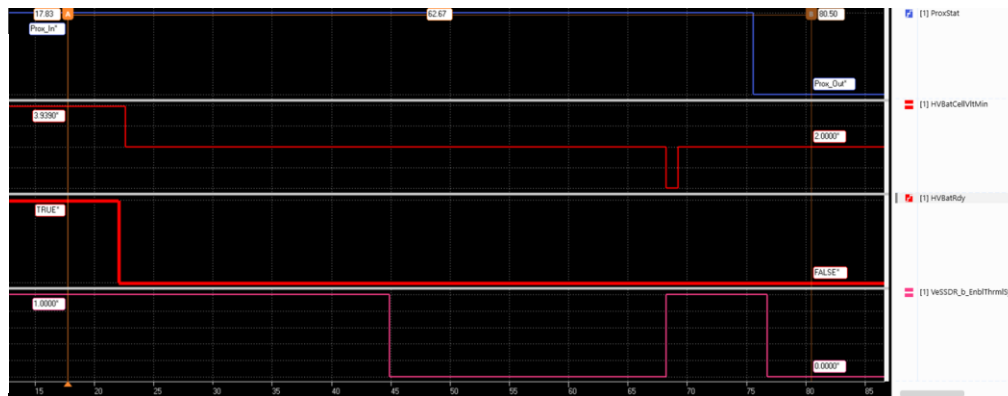
CAPL		IG BLOCK	
ADV	DISADV	ADV	DISADV
• Possibility to change a signal independently from the type of frame it belongs • Flexibility and adaptability • Possibility to change signals in two different CAN in the same time • Wide range of programming (limited only by the coding skills of the testing engineer)	• Need to write a C++ script • Physical interruption of CAN bus is needed • BoB (Breakout Box) is needed	• Both easy and fast to configure and use • Doesn't interrupt the CAN (parallel configuration)	• Possibility to change only message on event (strong limitation on usage cases)



EXAMPLE N°1 – Vary signals value

15	CFTS066 6.1.4.1	6.1.4.1 Drive Ready Management System Drive Ready Management system shall set PwrModg_ThermEnab = False if it detects that the Traction battery is below the Under-Voltage Threshold (TracBat_UV_Actv = True) and it is not plugged-in, for any of the above conditions. (1) Driver Door is open (Door_Ajar_Sts = True) (2) Ignition state is Accessory, Run, or Start (CmdlgnSts = Acc, Run, or Start) (3) Remote Start is active (Rem_Key_Strt = True) ((4) Vehicle is plugged in (Valid_Plugged_In = True)) (5) Wakeup is due to periodic wakeup for 12V charging (PerWU12VChrg = True) (6) Wakeup is due to periodic wakeup for Traction Battery conditioning (PerWU_Thermal_Condn = True) (7) Wakeup is due to scheduled charging (WakeUp_Sch_Charge = True)	HSC Wakeup - Plugged In Pre_Condition: 1. Make sure that the vehicle is in sleep (CAN Network down and CAN communication inactive) Test_Condition: 1. Plug in the vehicle in the EVSE connector. (OBCM_J1772_Data.ProxStat = 2 (Prox In)) 2. Set under voltage conditions for Traction battery using CANoe Gateway tool. (BPCM_MSG_01.HVBatCellVltMin < BPCM_MSG_04.HVBatCell_Voltage_Low_Thrsh) 3. Set HVBatRdy = 0 4. Plug out the vehicle. (OBCM_J1772_Data.ProxStat = 1 (Prox Out)) 5. Key OFF	P1P2/P1P4 Check the following: 4. After plugging out : VeSSDR_b_EnbThrmSys=0 (Observe voltage at pin B62/H358 = 0 V)
----	-----------------	---	--	--

```
else if(this.id==0x220 & X==4)
{
    message CAN1.BPCM_MSG_02 msg;
    msg.HVBatCellVltMin=2000;
    output(msg);
    msg = this;
    return;
}
else if(this.id==0x150 & X==4)
{
    message CAN1.BPCM_MSG_01 msg;
    msg.HVBatRdy=0;
    msg.CRC_150h=CRCCalc(msg);
    output(msg);
    msg = this;
    return;
}
```





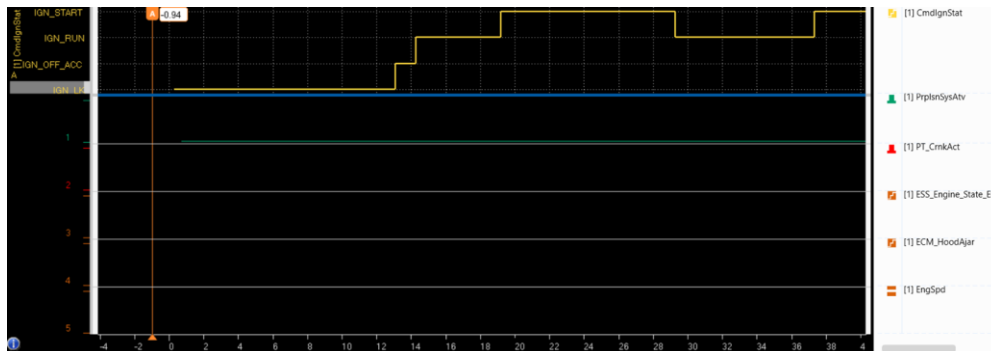
EXAMPLE N°2 – Set a Loss of Communication (LOC)

42	CFTS066 6.3.4.3	The Engine control system shall send the crank indication (ESS_Eng_PTCrnkActv) to Stop/Start Management system of HSC only when the following conditions are True: ☑ Crank is active ☑ There are no engine faults ☑ There are no immobilizer faults ☑ HCP is not sending the Crank Abort request	PTCrnkActv Engine Faults Pre_Condition: 1. Make sure that the vehicle is in sleep. (CAN Network down and CAN communication inactive) Test_Description: 1. Set engine faults (ECM loses communication with HCP) 2. Crank the vehicle 3. Verify acceptance criteria 4. Key OFF	Check the following: P1P2 2. (CBC_PT2.CmdIgnStat / HCP_GW_20.CmdIgnStat = 5 (Crank) 3. While Cranking: Engine_Torque_Status.PT_CrnkActv = 0 P1P4 2. (BODY2.CmdIgnSts / HCP_GW_20.CmdIgnStat = 5 (Crank) 3. While Cranking: Engine_Torque_Status.PT_CrnkActv = 0
----	-----------------	---	--	---

```

if ((this.id==0xA7 | this.id==0X101 | this.id==0X18D | this.id==0X192 | this.id==0X190 |
    this.id==0X18F | this.id==0X303 | this.id==0X18C | this.id==0X302 | this.id==0X310 |
    this.id==0X351 | this.id==0X35A | this.id==0X18B | this.id==0X18E | this.id==0X191 |
    this.id==0XAA | this.id==0XAB ) & X ==1)
{
    message CAN1.* msg;
    //output(msg);
    msg = this;
    return;
}

```





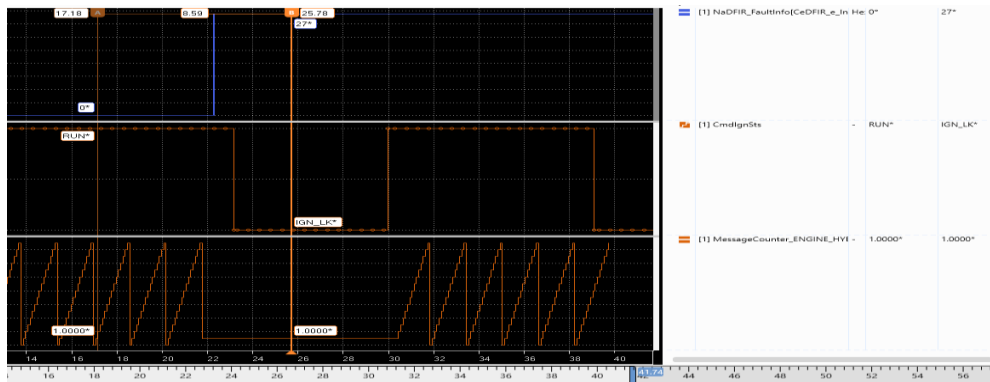
```
if(this.id==0x40B & X==1) // ENGINE_HYBD_FD_3 on key 1
{
    message CAN1.ENGINE_HYBD_FD_3 msg;
    CRCposC1=23;
    msg.CRC_ENGINE_HYBD_FD_3=0;
    output(msg);
    msg = this;
    return;
}
```

U0401- Implausible Data Received From ECM/PCM "A" is set



```
if(this.id==0x40b & X==1)    // ENGINE_HYBD_FD_3 on key 1 to modify MC
{
    message CAN1.ENGINE_HYBD_FD_3 msg;
    msg.MessageCounter_ENGINE_HYBD_FD_3=1;
    CRCposC1=23;
    msg.CRC_ENGINE_HYBD_FD_3=CRCCalc(msg);
    output(msg);
    msg = this;
    return;
}
```

Scope: set U0401- Implausible Data
Received From ECM/PCM "A"





EXERCISES

- Write a script in order to change a signal of a frame which contains CRC check
- Write a script in order to change more signals of a frame which contains CRC check
- Write a script in order to set implausible data through CRC corruption (i.e. sum number 10 to the CRC value calculated by the dedicated function)
- Write a script in order to change two signals which belong to different messages
- Write a script in order to change two signals which belong to different messages; the second signal shall be changed after the first signal is modified (e.g. consider a time lap of some seconds between the two modifications)
- Try to send a diagnostic frame, monitor the request and the response in CANalyzer

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